

OE-EC804A — Artificial Intelligence

Semester VIII • 3L:0T:0P • 3 Credits

Module 1: Introduction to Artificial Intelligence & Intelligent Agents

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Overview and Foundations of AI

Artificial Intelligence (AI) is the study and design of intelligent agents capable of perceiving their environment, reasoning, learning, and taking actions to achieve specific goals.

Definition of Artificial Intelligence

Artificial Intelligence (AI) is a branch of computer science concerned with building software and systems that perform tasks requiring human-like cognitive abilities, such as learning, reasoning, problem-solving, perception, and decision-making.

1.1 The Four Definitions of AI

AI definitions are historically categorized along two dimensions: **Human Performance vs. Rationality** and **Thinking vs. Acting**.

Dimension	Thinking Focus	Acting Focus
Human-Centric	Thinking Humanly: Cognitive modeling; validating AI models against human brain activity.	Acting Humanly: The Turing Test approach; capability to fool an interrogator.
Rationality-Centric	Thinking Rationally: "Laws of Thought" approach; syllogisms and formal logic.	Acting Rationally: Rational Agent approach; acting to achieve the best expected outcome.

1.2 Foundations of AI

AI is an interdisciplinary field built on insights from multiple disciplines:

- **Philosophy:** Formal logic, theories of mind, epistemology, and nature of knowledge.
- **Mathematics:** Probability, formal logic, statistics, linear algebra, graph theory, and computability theory.
- **Economics:** Decision theory, game theory, utility theory, and operations research.
- **Neuroscience:** Biological neural networks, brain structure, processing mechanisms.
- **Psychology / Cognitive Science:** Human perception, memory models, cognitive architectures.
- **Control Theory & Cybernetics:** Self-regulating feedback systems, optimal control.

1.3 History & Evolution of AI

- **1943–1955 (Gestation):** McCulloch & Pitts artificial neuron model; Turing's 1950 paper "Computing Machinery and Intelligence".
- **1956 (Birth of AI):** Dartmouth Conference organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon; coining of the term "Artificial Intelligence".
- **1966–1974 (Early Enthusiasm):** Logic Theorist, ELIZA, Microworlds (BLOCKSWORLD).
- **1974–1980 (First AI Winter):** Realization of computational complexity limits (NP-completeness).

- **1980–1987 (Expert Systems Era):** MYCIN, XCON (R1), knowledge-based rule systems.
- **1987–1993 (Second AI Winter):** Collapse of specialized LISP machine hardware market.
- **1993–Present (Modern Era):** Probabilistic reasoning, Big Data, Deep Learning, and Large Language Models (LLMs).

1.4 The Turing Test

Proposed by Alan Turing in 1950 ("Imitation Game"), it evaluates whether a computer can demonstrate human-equivalent intelligence.

Capabilities Required for the Turing Test

To pass the Total Turing Test, an AI system requires:

1. **Natural Language Processing (NLP):** To communicate effectively in human language.
2. **Knowledge Representation:** To store information and concepts.
3. **Automated Reasoning:** To use stored knowledge to answer questions and draw conclusions.
4. **Machine Learning:** To adapt to new circumstances and detect patterns.
5. **Computer Vision:** To perceive objects and physical inputs (Total Turing Test).
6. **Robotics:** To manipulate objects and move in physical space (Total Turing Test).

2. Intelligent Agents and Environments

Definition of an Agent

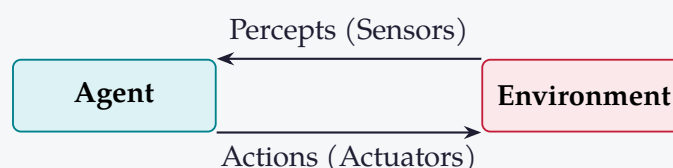
An **Agent** is anything that perceives its environment through **sensors** and acts upon that environment through **actuators**. The mapping from percept sequences to actions is called the **Agent Function**.

Mathematical Formulation of an Agent

$$f : \mathcal{P}^* \rightarrow \mathcal{A} \quad (1)$$

where $\mathcal{P}^*$ is the sequence of all past percepts, and $\mathcal{A}$ is the set of actions available to the agent.

Agent-Environment Interaction Loop



2.1 PEAS Specification

To design an intelligent agent, we must explicitly define its **PEAS** properties:

- **P — Performance Measure:** Objective criteria evaluating the success of the agent's behavior.
- **E — Environment:** External surroundings and domain in which the agent operates.
- **A — Actuators:** Mechanisms through which the agent performs actions.
- **S — Sensors:** Instruments through which the agent receives inputs/percepts.

Agent Type	Performance Measure	Environment	Actuators	Sensors
Automated Taxi Driver	Safe, fast, legal, comfortable, max profit	Roads, traffic, pedestrians, weather	Steering, accelerator, brake, horn	Cameras, LiDAR, GPS, speedometer
Medical Diagnosis	Healthy patient, min cost, no lawsuits	Patient, hospital staff, laboratory	Screen display (reports, prescriptions)	Keyboard (symptoms, lab test results)
Vacuum Cleaner	Clean floor, min time, min electricity	Rooms, dirt, obstacles, carpet	Wheels, vacuum motor, brushes	Bump sensor, dirt detection sensor

2.2 Concept of Rationality

A **Rational Agent** is one that operates to choose actions that maximize its expected performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent possesses.

Factors Determining Rationality

Rationality depends on 4 parameters:

1. The **Performance Measure** defining the success criterion.
2. The agent's **Prior Knowledge** of the environment.
3. The **Actions** that the agent can perform.
4. The agent's **Percept Sequence** up to the present moment.

3. Environment Properties and Agent Structures

3.1 Classification of Environments

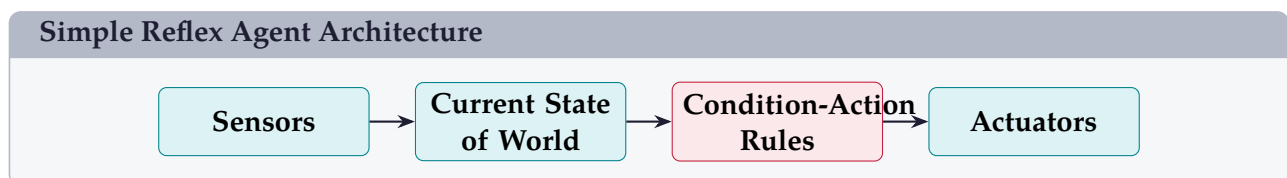
- **Fully Observable vs. Partially Observable:** An environment is fully observable if sensors give complete state access at any time (e.g., Chess vs. Poker / Medical Diagnosis).
- **Single-Agent vs. Multi-Agent:** Single-agent operates alone (Crossword); Multi-agent involves other agents (Chess, Taxi driving). Multi-agent can be competitive or cooperative.
- **Deterministic vs. Stochastic:** Deterministic if next state is completely determined by current state and action (Chess). Stochastic if uncertainty exists (Taxi driving, Ludo).

- **Episodic vs. Sequential:** Episodic means current decision does not affect future decisions (Image classification). Sequential means current actions affect future decisions (Chess).
- **Static vs. Dynamic:** Static environment does not change while agent is deliberating (Crossword). Dynamic environment changes continuously (Taxi driver). Semi-dynamic if environment doesn't change but agent score decreases with time (Timed Chess).
- **Discrete vs. Continuous:** Discrete has finite, distinct states and actions (Chess). Continuous has continuous parameters like speed, angle, temperature (Taxi driver).
- **Known vs. Unknown:** Known means environment laws/physics are given to the agent (Solitaire). Unknown means laws must be learned (New video game).

3.2 Four Basic Agent Architectures

3.2.1 1. Simple Reflex Agent

Selects actions based **only on the current percept**, ignoring percept history. Uses Condition-Action Rules ('IF condition THEN action').



3.2.2 2. Model-Based Reflex Agent

Maintains an internal **state** tracking aspects of the environment not currently visible. Requires knowledge of how the world evolves and how agent actions affect the world.

3.2.3 3. Goal-Based Agent

Combines state information with **goal information** to choose actions that achieve the desired outcome. Uses search and planning.

3.2.4 4. Utility-Based Agent

Uses a **utility function** mapping a state onto a real number to measure happiness/desirability of states. Allows tradeoff choices under conflicting goals.

Quick Revision — Module 1

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Key Concepts and Metrics at a Glance

Agent Function	$f : \mathcal{P}^* \rightarrow \mathcal{A}$ mapping percept histories to actions.
PEAS Specification	Performance Measure, Environment, Actuators, Sensors.
Rationality	Maximizes expected performance based on percept sequence & prior knowledge.
Turing Test	6 core capabilities: NLP, Knowledge Rep, Reasoning, ML, Vision, Robotics.
Environment Dimensions	7 attributes: Observable, Multi-agent, Deterministic, Sequential, Static, Discrete, Known.

One-Line Definitions

- **Rational Agent:** An agent that acts to achieve the best expected performance outcome.
- **PEAS:** Formal framework defining an agent's task environment.
- **Simple Reflex Agent:** Operates strictly on current percept using Condition-Action rules.
- **Model-Based Agent:** Maintains internal state to handle partial observability.
- **Utility-Based Agent:** Evaluates action choices using a quantitative utility function.

Frequently Asked / Viva Questions

Q1. Define Artificial Intelligence. List the four perspectives of AI definitions.

AI is the field of computer science creating systems capable of human-like intelligence. The 4 perspectives are: Thinking Humanly, Acting Humanly (Turing Test), Thinking Rationally (Logic), and Acting Rationally (Rational Agents).

Q2. What is the Turing Test? What capabilities are required to pass it?

Proposed by Alan Turing in 1950, it tests if a machine can imitate human conversation convincingly. Required capabilities: NLP, Knowledge Representation, Automated Reasoning, Machine Learning, Computer Vision, and Robotics.

Q3. What is an Agent and Agent Function?

An agent perceives environment via sensors and acts via actuators. Agent function $f : \mathcal{P}^* \rightarrow \mathcal{A}$ maps percept histories to actions.

Q4. Explain the PEAS framework with an example.

PEAS stands for Performance, Environment, Actuators, Sensors. For automated

Frequently Asked / Viva Questions (contd.)

taxi: P=Safety/Speed, E=Roads/Traffic, A=Steering/Brake, S=Cameras/Radar.

Q5. Distinguish between Rationality and Omniscience.

Rationality maximizes expected performance based on available percepts. Omniscience knows actual outcome in advance. Rationality is feasible; omniscience is impossible in uncertain environments.

Q6. Differentiate between Deterministic and Stochastic environments.

Deterministic: Next state is completely fixed by current state and action.
Stochastic: Next state involves probability and randomness.

Q7. What is a Simple Reflex Agent?

An agent that acts based strictly on the current percept using IF-THEN condition-action rules without memory.

Q8. Why does a Model-Based Reflex Agent need internal state?

To handle partially observable environments by tracking unobserved aspects of the world over time.

Q9. Compare Goal-Based vs Utility-Based Agents.

Goal-based agents focus on reaching a target state (binary success). Utility-based agents evaluate degree of desirability (continuous score/utility) to make optimal tradeoffs.

Q10. Explain Episodic vs Sequential environments.

Episodic: Each episode is independent; action choice does not impact future states. Sequential: Actions impact all future states (e.g. Chess).